



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-G2

SYSTEMS INTEGRATION AND

ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS

SPECIFICATION (FRS)

Project Title: Mini App – User Registration & Authentication

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1. Introduction

1.1. Purpose

The purpose of this system is to provide a simple User Registration and Authentication for a web-based mini application.

1.2. Scope

The System will allow:

- New users to Register an account with a Username, Email and Password.
- Registered users to Login using their credentials
- Logged in users to Access the Dashboard, their Profile Page and Protected Pages.
- Users to Logout preventing Access to protected Pages.

System Boundaries:

- The System will only focus on User Authentication, Dashboard Access, Profile Page Access and Protected Page Access.
- The System will not include Advanced features such as Multi-factor Authentication and API based token Authentication.

1.3. Definitions, Acronyms, and Abbreviations

Term	Definition
User	A person who interacts with the System that is either a guest or a registered user.
Dashboard	The page where logged in users will be redirected to upon successful logins. And where users can view personalized information.
Session	This is a Temporary storage that is used to maintain a user's login state.

2. Overall Description

2.1. System Perspective

The system will be a standalone mini web application that provides a simple Authentication workflow. This can be integrated into larger applications in the future.

2.2. User Classes and Characteristics

User Type	Characteristics	Actions
Guest User	Not registered or not logged-in	Register, login
Authenticated User	Successfully logged-in	Access dashboard, profile, protected pages and logout.

2.3. Operating Environment

Hardware: Desktop or Laptop.

Software:

- Web browser
- ReactJS front end and Spring boot backend
- Database: MySQL

Tools:

Code editor (IDE): VS Code

2.4. Assumptions and Dependencies

- The system assumes that the database is operational and accessible.
- Session-based authentication is sufficient for user login and logout.

3. System Features and Functional Requirements

3.1. Feature 1:

Description: Allows new users to Create and Account by providing a Username, Email and Password.

Functional Requirements:

- The system must validate that all input fields are filled.
- The system must check that the email inputted is unique.
- Upon successful registration, the user must be redirected to the Login Page.

3.2. Feature 2:

Description: Allows Registered users to log in with their Email and Password.

Functional Requirements:

- logged in users are redirected to the dashboard
- The system must validate user's credentials against the stored data in the database
- if credentials are correct, then the system creates a session for the user.
- if credentials are invalid, then the system must show an error message.

4. Non-Functional Requirements

Performance: The system must respond fast and not lag. It must respond to Login and Registration requests within a short time.

Security: Passwords must be Hashed and User sessions must be protected.

Usability: The UI/UX must be Clean and Simple.

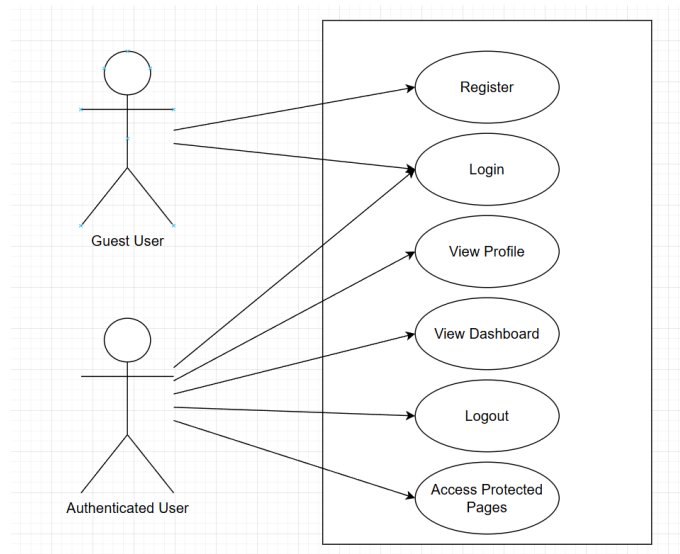
Reliability: The System must be able to handle multiple current users without errors.

5. System Models (Diagrams)

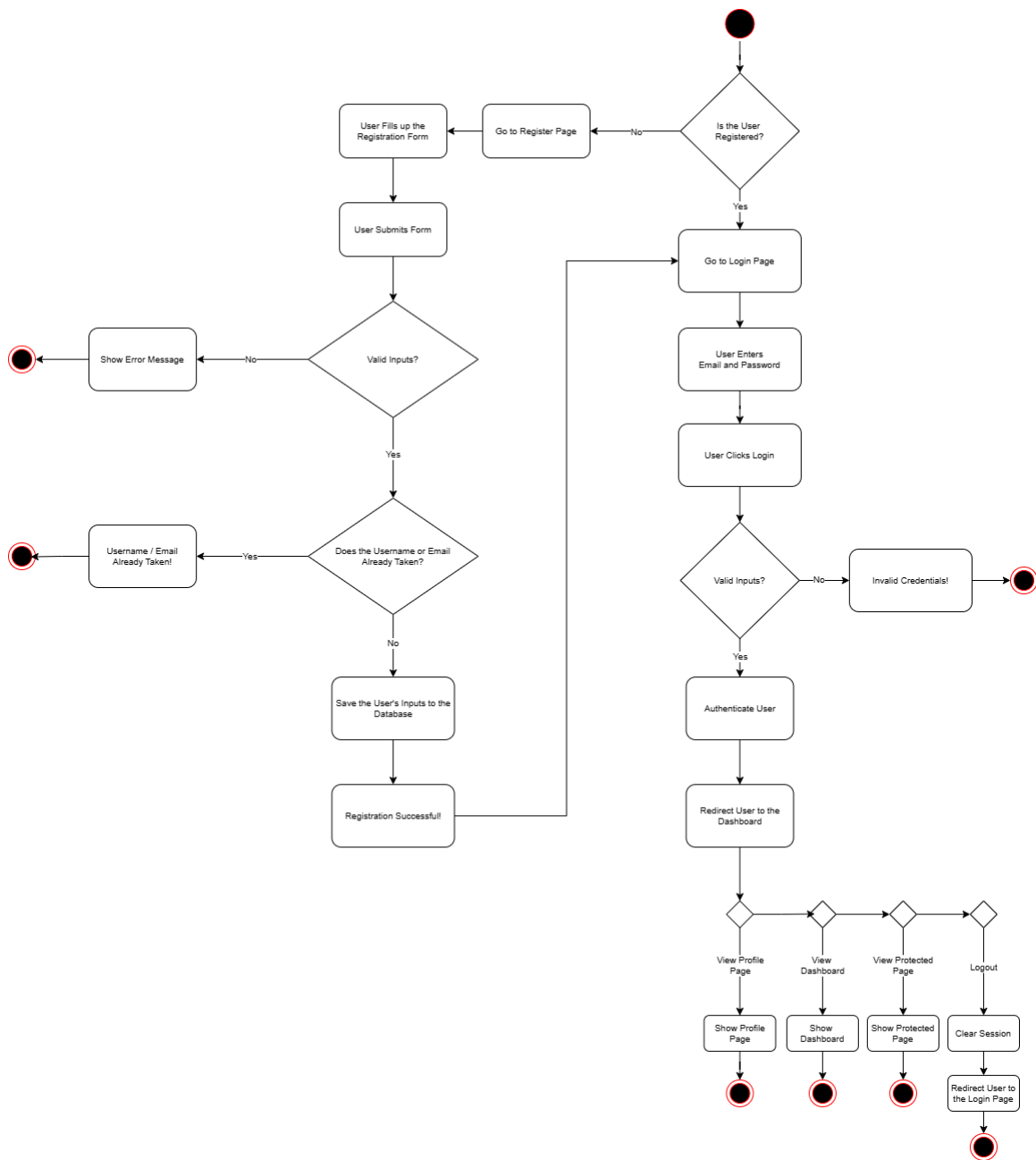
5.1. ERD

User		
PK	<u>userID</u>	<u>int</u>
	username	varchar(255)
	email	varchar(255)
	password	varchar(255)
	dateCreated	date

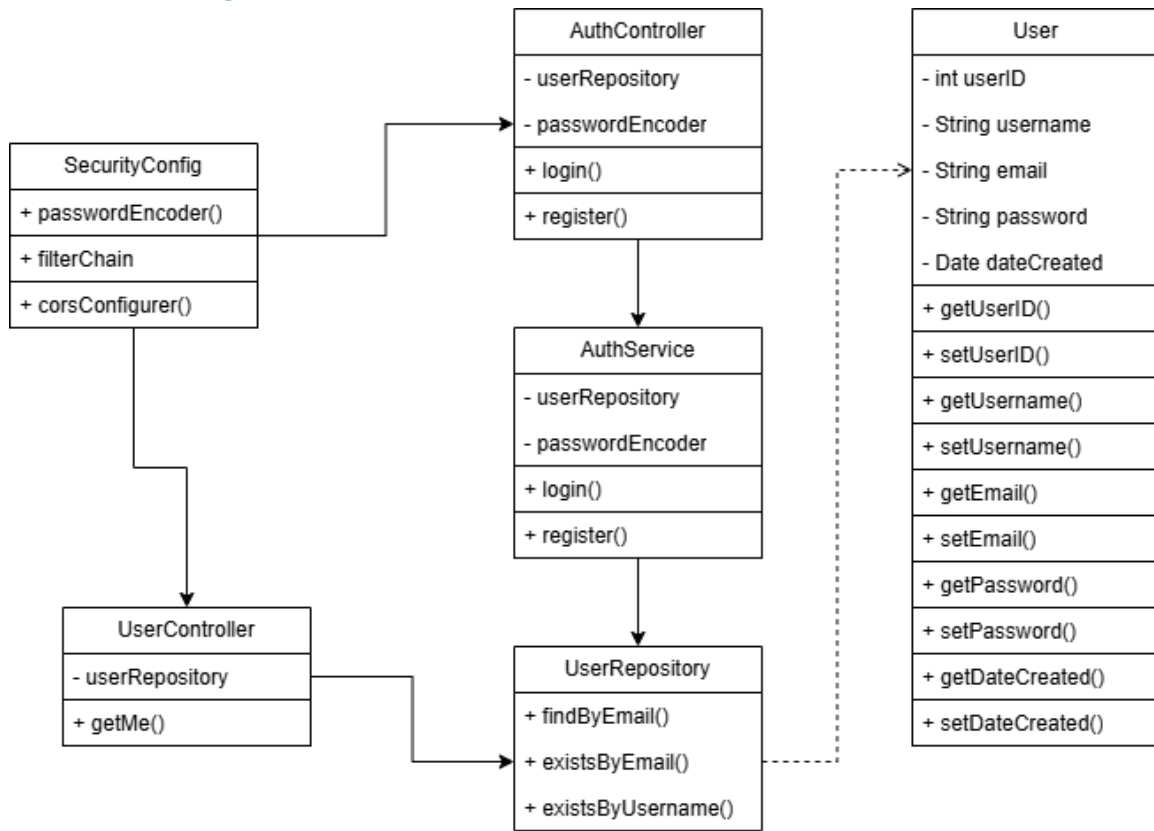
5.2. Use Case Diagram



5.3. Activity Diagram

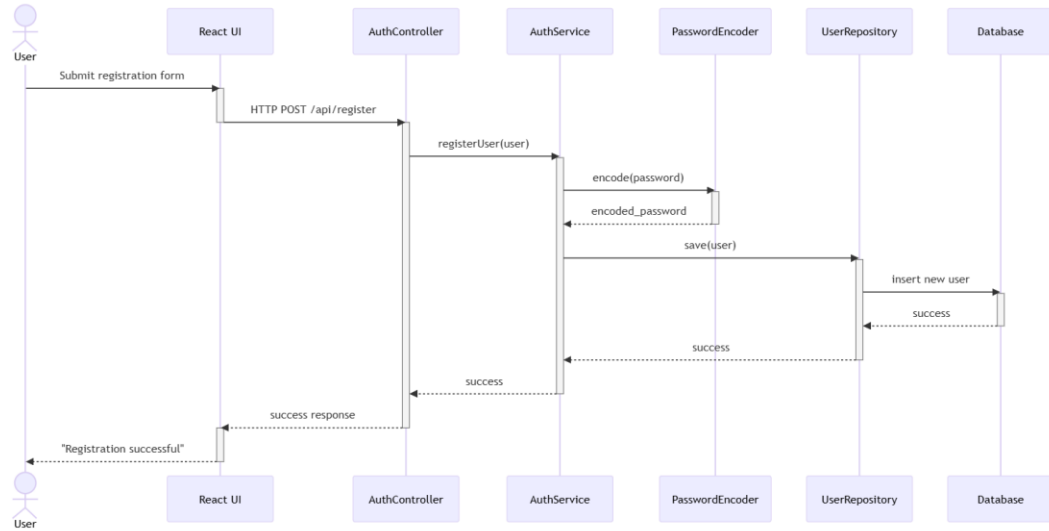


5.4. Class Diagram

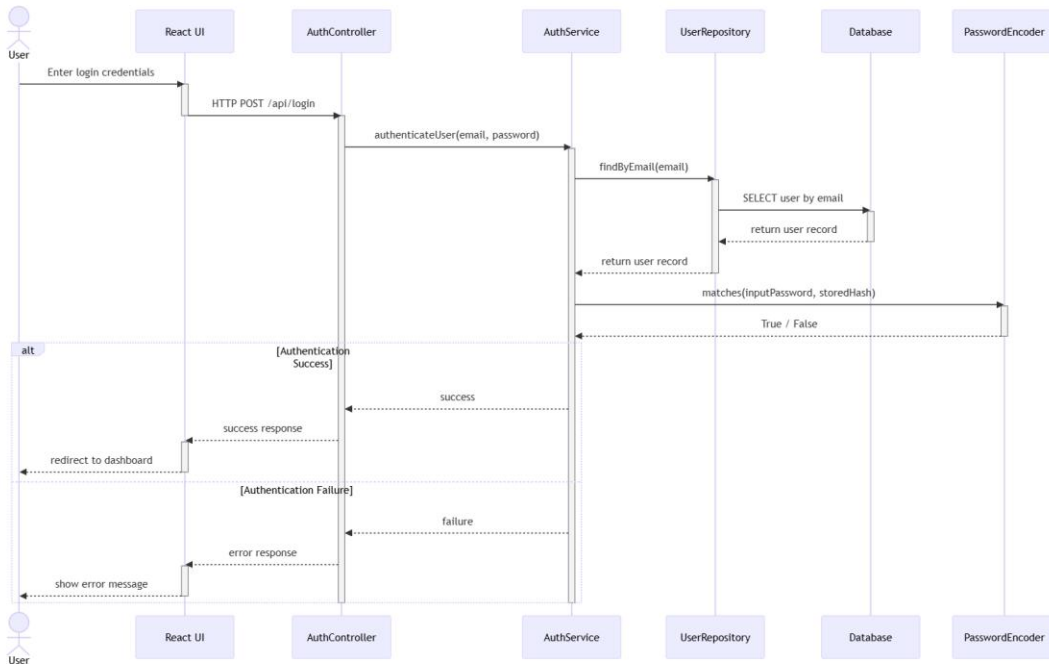


5.5. Sequence Diagram

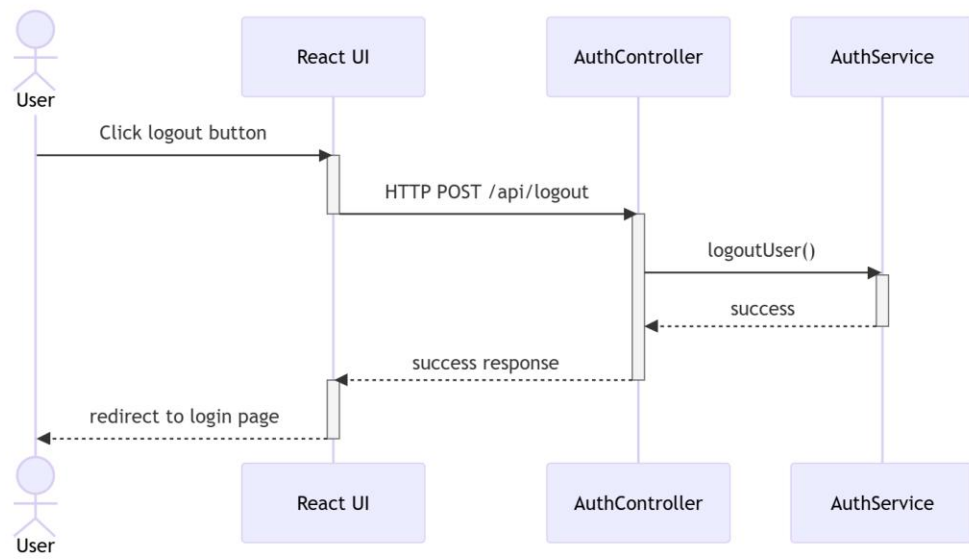
Registration Sequence:



Login Sequence:

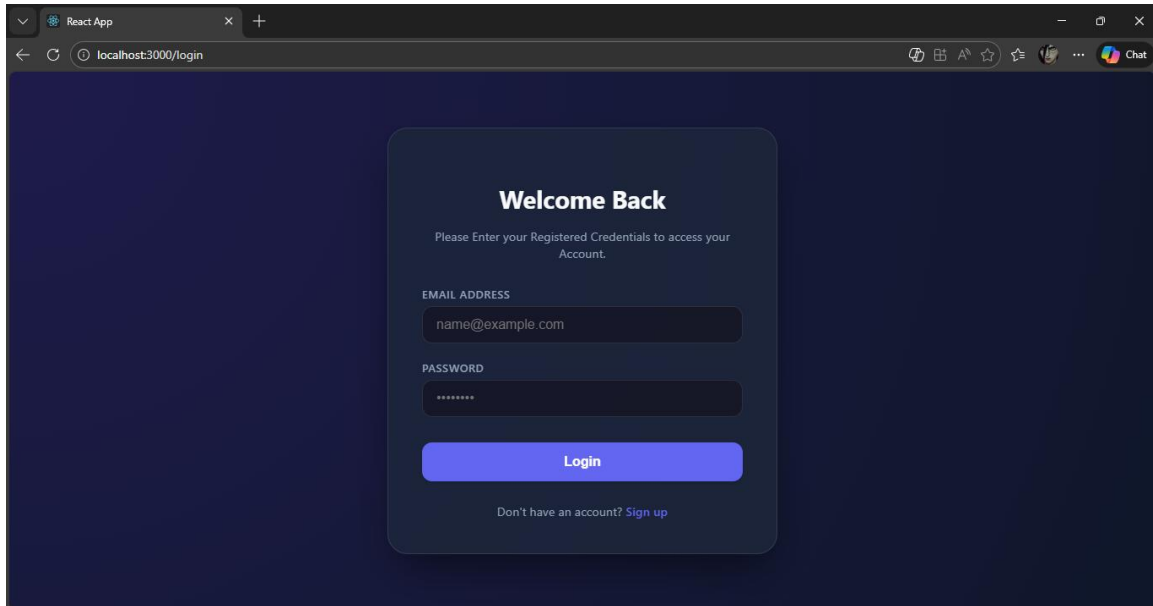


Logout Sequence:



Web UI Screenshots

Login Page:



A screenshot of a web browser displaying the login page of a React App. The browser's address bar shows 'localhost:3000/login'. The page has a dark blue background. In the center, there is a light blue rounded rectangle containing the following elements: a heading 'Welcome Back', a subtext 'Please Enter your Registered Credentials to access your Account.', an 'EMAIL ADDRESS' label above a text input field containing 'name@example.com', a 'PASSWORD' label above a password input field with masked characters, a blue 'Login' button, and a link 'Don't have an account? Sign up'.

Welcome Back

Please Enter your Registered Credentials to access your Account.

EMAIL ADDRESS

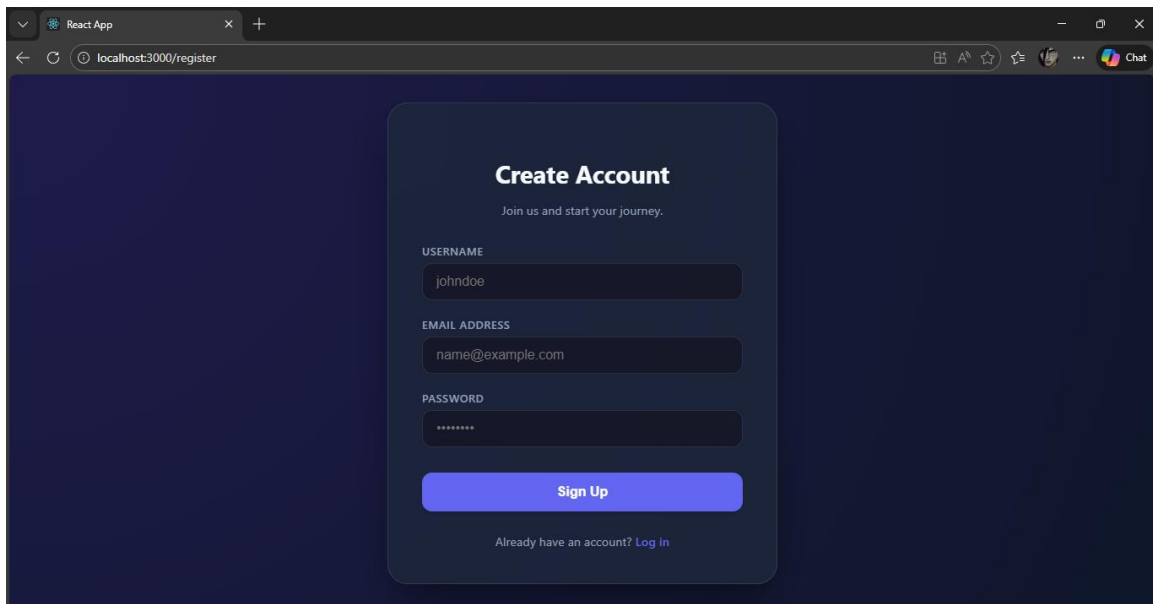
name@example.com

PASSWORD

Login

Don't have an account? [Sign up](#)

Register Page:



A screenshot of a web browser displaying the register page of a React App. The browser's address bar shows 'localhost:3000/register'. The page has a dark blue background. In the center, there is a light blue rounded rectangle containing the following elements: a heading 'Create Account', a subtext 'Join us and start your journey.', a 'USERNAME' label above a text input field containing 'johndoe', an 'EMAIL ADDRESS' label above a text input field containing 'name@example.com', a 'PASSWORD' label above a password input field with masked characters, a blue 'Sign Up' button, and a link 'Already have an account? Log in'.

Create Account

Join us and start your journey.

USERNAME

johndoe

EMAIL ADDRESS

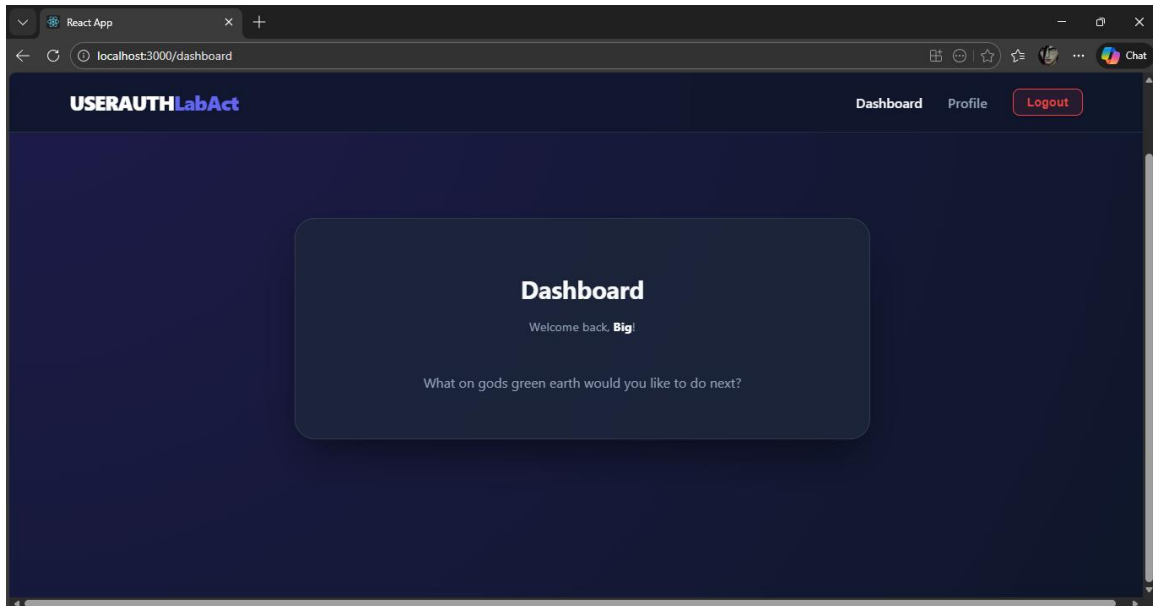
name@example.com

PASSWORD

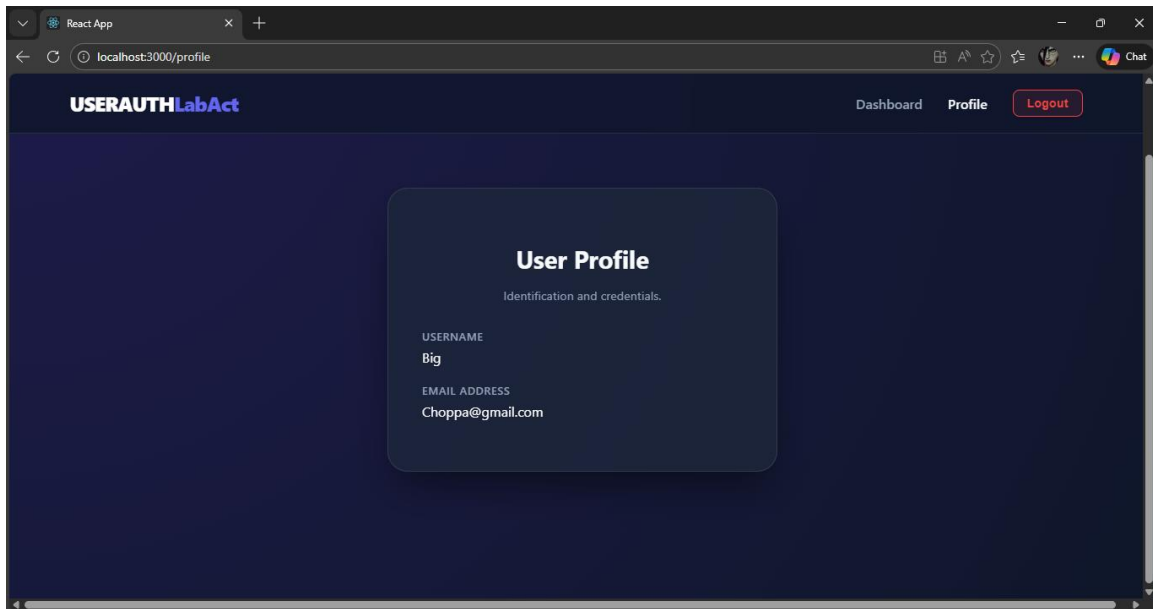
Sign Up

Already have an account? [Log in](#)

Dashboard:



Profile Page:



6. Appendices

1. Lucidchart. UML Sequence Diagram Guide
<https://www.lucidchart.com/pages/uml-sequence-diagram>
2. TechTarget. Sequence Diagram Definition
<https://www.techtarget.com/searchsoftwarequality/definition/sequence-diagram>
3. Visual Paradigm. Sequence Diagram Tutorial
<https://online.visual-paradigm.com/diagrams/tutorials/sequence-diagram-tutorial/>
4. Research on Functional vs Non-Functional Requirements
arxiv.org/abs/2211.05286